HW 2 - Zeshui Song

3.21

21. Determine the value of v_x as labeled in the circuit of Fig. 3.62.

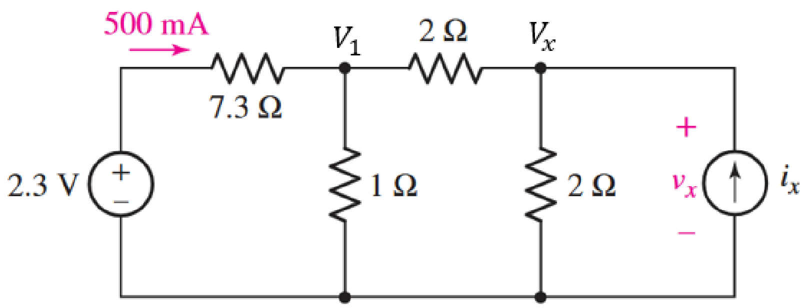


FIGURE 3.62

Using nodal analysis, assigning V_1 and V_x, I can apply KCL at node 1:

$$[500 \times 10^{-3}\,\mathrm{A}] = \frac{V_1}{[1\,\Omega]} + \frac{V_1 - V_x}{[2\,\Omega]}$$

Using KVL (CCW) starting from negative terminal of the voltage source around the left loop to find V_1:

$$[2.3\,\mathrm{V}] - [500 \times 10^{-3}\,\mathrm{A}][7.3\,\Omega] - V_1 = 0$$

$$V_1 = -1.35\,\mathrm{V}$$

Plugging in V_1 into the KCL equation to find V_x:

$$[500 \times 10^{-3}\,\mathrm{A}] = \frac{[-1.35\,\mathrm{V}]}{[1\,\Omega]} + \frac{[-1.35\,\mathrm{V}] - V_x}{[2\,\Omega]}$$

$$\boxed{V_x = -5.05\,\mathrm{V}}$$

3.23

23. (*a*) Determine a numerical value for each current and voltage (i_1, v_1, etc.) in the circuit of Fig. 3.64. (*b*) Calculate the power absorbed by each element and verify that they sum to zero.

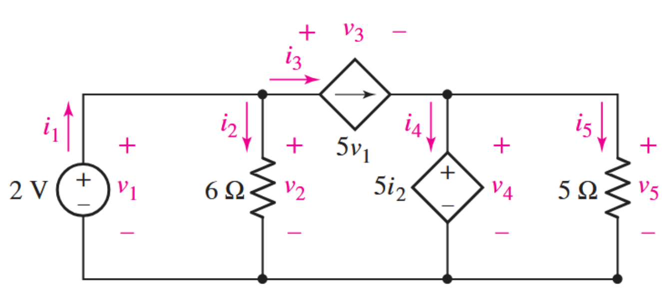


FIGURE 3.64

(a)

i_2 and v_2: KVL (CCW) starting from negative terminal of the 2V source around the left loop:

$$[2\,\text{V}] - i_2[6\,\Omega] = 0 \quad \implies i_2 = 0.33\,\text{A}$$
$$[2\,\text{V}] - v_2 = 0 \quad \implies v_2 = 2\,\text{V}$$

i_3 and v_4 by plugging in for v_1 and i_2:

$$i_3 = 5v_1 = 5[2\,\text{V}] = 10\,\text{A}$$

$$v_4 = 5i_2 = 5[0.33\,\text{A}] = 1.67\,\text{V}$$

i_1 by KCL at top node of $6\,\Omega$ resistor:

$$i_1 = i_2 + i_3 = [0.33\,\text{A}] + [10\,\text{A}] = 10.33\,\text{A}$$

v_3 by KVL (CCW) starting from negative terminal of the $6\,\Omega$ resistor around the middle loop:

$$v_2 - v_3 - v_4 = 0 \quad \implies v_3 = [2\,\text{V}] - [1.67\,\text{V}] = 0.33\,\text{V}$$

v_5 is in parallel with v_4, thus it has the same voltage:

$$v_5 = v_4 = 1.67\,\text{V}$$

i_5 by Ohm's law:

$$i_5 = \frac{[1.67\,\text{V}]}{[5\,\Omega]} = 0.334\,\text{A}$$

Lastly, i_4 by KCL at top node of v_4:

$$i_3 = i_4 + i_5 \quad \implies i_4 = i_3 - i_5 = [10\,\text{A}] - [0.334\,\text{A}] = 9.666\,\text{A}$$

Element	**Voltage, v [V]**	**Current, i [A]**
2V Source	$v_1 = 2$	$i_1 = 10.33$
6 Ω Resistor	$v_2 = 2$	$i_2 = 0.33$
Dependent $5v_1$	$v_3 = 0.33$	$i_3 = 10$
Dependent $5i_2$	$v_4 = 1.67$	$i_4 = 9.66$
5 Ω Resistor	$v_5 = 1.67$	$i_5 = 0.33$

(b)

Element	**Voltage,** v **[V]**	**Current,** i **[A]**	$P = Vi$ **[W]**
2V Source	$v_1 = 2$	$i_1 = 10.33$	$P_1 = -20.66$
6 Ω Resistor	$v_2 = 2$	$i_2 = 0.33$	$P_2 = 0.66$
Dependent $5v_1$	$v_3 = 0.33$	$i_3 = 10$	$P_3 = 3.30$
Dependent $5i_2$	$v_4 = 1.67$	$i_4 = 9.66$	$P_4 = 16.13$
5 Ω Resistor	$v_5 = 1.67$	$i_5 = 0.33$	$P_5 = 0.55$
Total Power Absorbed (Not exactly 0 b/c of rounding)			**-0.02**

Note: The sign convention for power *absorbed* is negative for sources (current enters - and leaves + terminal) and positive for loads (current enters + and leaves - terminal).

3.34

34. Although drawn so that it may not appear obvious at first glance, the circuit of Fig. 3.74 is in fact a single-node-pair circuit. (*a*) Determine the power absorbed by each resistor. (*b*) Determine the power supplied by each current source. (*c*) Show that the sum of the absorbed power calculated in (*a*) is equal to the sum of the supplied power calculated in (*b*).

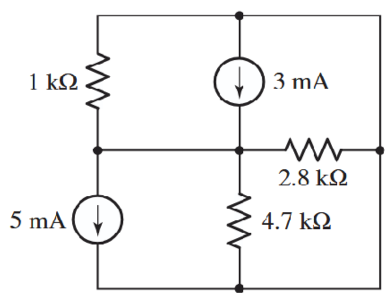
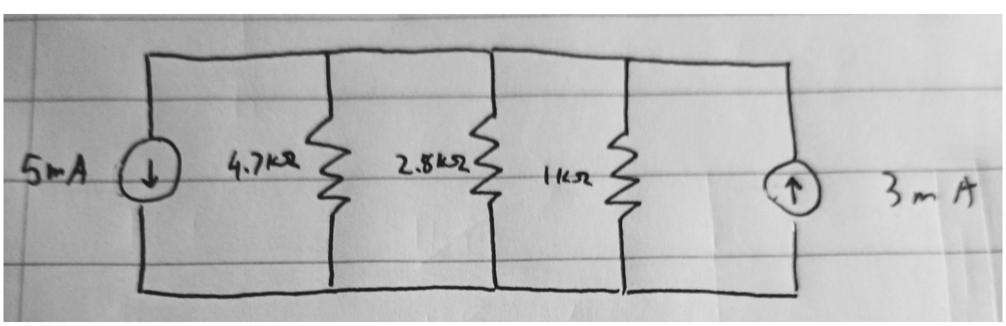


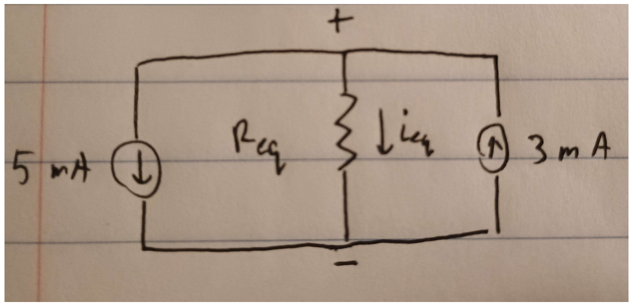
FIGURE 3.74

This is a **single-node pair circuit** meaning that every element is in parallel across the same two nodes. Thus, it can be redrawn:



Note: The direction of the current sources has to be consistent with the direction in which they feed into/out of the resistors. **(a)** We can further simplify by combining the resistors in parallel:

$$\frac{1}{R_{eq}} = \frac{1}{[4.7 \times 10^3\,\Omega]} + \frac{1}{[2.8 \times 10^3\,\Omega]} + \frac{1}{[1 \times 10^3\,\Omega]} \quad \Longrightarrow \quad R_{eq} = 636.979\,\Omega$$



Using KCL at the top node:

$$i_{\text{eq}} = [3 \times 10^{-3}\,\text{A}] - [5 \times 10^{-3}\,\text{A}] = -2 \times 10^{-3}\,\text{A}$$

By Ohm's law:

$$V = [(-2 \times 10^{-3}\,\text{A})][636.979\,\Omega] = -1.2739\,\text{V}$$

Element	**Voltage, v [V]**	**Current, $i = V/R$ [A]**	$P = vi$ **[W]**
5 mA Source	-1.2739	5×10^{-3}	-6.369×10^{-3}
4.7 kΩ Resistor	-1.2739	-0.271×10^{-3}	$.345 \times 10^{-3}$
2.8 kΩ Resistor	-1.2739	-0.454×10^{-3}	0.578×10^{-3}
1 kΩ Resistor	-1.2739	-1.273×10^{-3}	1.621×10^{-3}
3 mA Source	-1.2739	-3×10^{-3}	3.821×10^{-3}
Total Power Absorbed (Not exactly 0 b/c of rounding)			**-0.004**

Note: I had previously defined the positive voltage and current to be top to down, which is why everything turned out negative, except for the current from the 5 mA source.

3.38

38. Determine the value of v_1 required to obtain a zero value for the current labeled i in the circuit of Fig. 3.77.

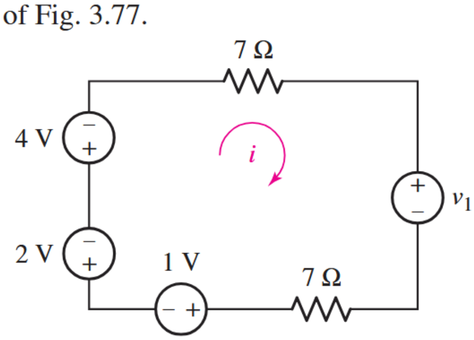


FIGURE 3.77

Using KVL (CCW) starting from the positive terminal of the 2 V source around the loop:

$$-[2\,\text{V}] - [4\,\text{V}] - i \cdot [7\,\Omega] - v_1 - i \cdot [7\,\Omega] - [1\,\text{V}] = 0$$

Since $i = 0$, we have:

$$-[2\,\mathrm{V}] - [4\,\mathrm{V}] - v_1 - [1\,\mathrm{V}] = 0 \qquad \Longrightarrow \qquad \boxed{v_1 = -7\,\mathrm{V}}$$

3.39

39. (*a*) For the circuit of Fig. 3.78, determine the value for the voltage labeled *v*, after first simplifying the circuit to a single current source in parallel with two resistors. (*b*) Verify that the power supplied by your equivalent source is equal to the sum of the supplied powers of the individual sources in the original circuit.

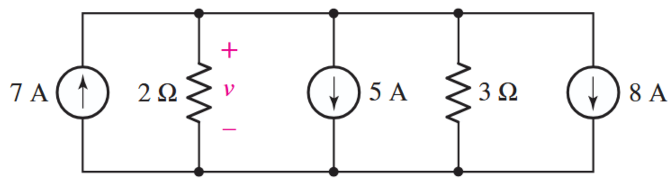


FIGURE 3.78

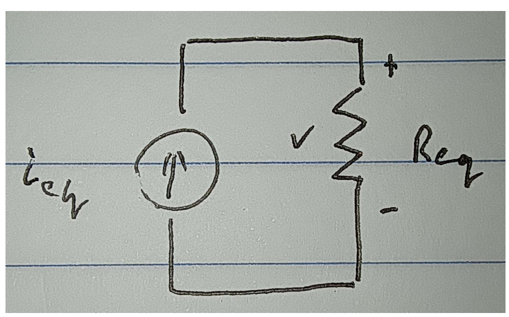
(a) Similar to problem *3.34*, if I combine the resistors in parallel, I can use KCL at the top node to find i_{eq}, and thus an equivalent current source and resistor in parallel:

$$i_{\text{eq}} = [7\,\mathrm{A}] - [5\,\mathrm{A}] - [8\,\mathrm{A}] = -6\,\mathrm{A}$$

$$R_{\text{eq}} = \frac{[2\,\Omega]\cdot[3\,\Omega]}{[2\,\Omega]+[3\,\Omega]} = 1.2\,\Omega$$

Thus, the voltage across each element is:

$$v = [(-6\,\mathrm{A})][1.2\,\Omega] = \boxed{-7.2\,\mathrm{V}}$$



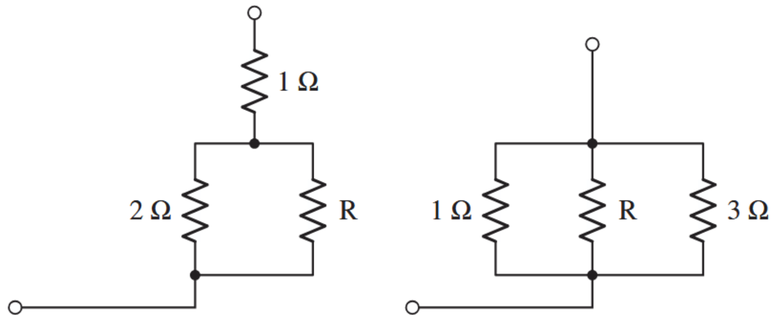
(b) Since the voltage v supplied by each current source is the same, the power supplied by each current source is then dependent only on the current supplied (following the sign convention for positive voltage and current):

$$P_{7\text{ A}} = [-7.2\,\mathrm{V}]\cdot[7\,\mathrm{A}] = -50.4\,\mathrm{W}$$
$$P_{5\text{ A}} = [-7.2\,\mathrm{V}]\cdot[-5\,\mathrm{A}] = 36\,\mathrm{W}$$
$$P_{8\text{ A}} = [-7.2\,\mathrm{V}]\cdot[-8\,\mathrm{A}] = 57.6\,\mathrm{W}$$
$$\Longrightarrow \boxed{\sum P = 43.2\,\mathrm{W}}$$

For the equivalent current source:

$$P_{\text{eq}} = [-7.2\,\mathrm{V}]\cdot[-6\,\mathrm{A}] = \boxed{43.2\,\mathrm{W}}$$

3.43 43. For each network depicted in Fig. 3.82, determine a single equivalent resistance if R = (*a*) 2 Ω; (*b*) 4 Ω; (*c*) 0 Ω.



(*a*) (*b*)

(a)
Combine the resistors in parallel:

$$R_{\|} = \frac{[2\,\Omega] \cdot R}{[2\,\Omega] + R}$$

Combine with the series resistor:

$$R_{\text{eq}} = [1\,\Omega] + R_{\|} = 1 + \frac{2R}{2+R}$$

(i) $R = 2\,\Omega$:

$$R_{\text{eq}} = 1 + \frac{2R}{2+R} = \boxed{2\,\Omega}$$

(ii) $R = 4\,\Omega$:

$$R_{\text{eq}} = 1 + \frac{2R}{2+R} = \boxed{2.33\,\Omega}$$

(iii) $R = 0\,\Omega$:

$$R_{\text{eq}} = 1 + \frac{2R}{2+R} = \boxed{1\,\Omega}$$

(b)
Combine the resistors in parallel:

$$\frac{1}{R_{\text{eq}}} = \frac{1}{[1\,\Omega]} + \frac{1}{[R\,\Omega]} + \frac{1}{[3\,\Omega]} \quad \Longrightarrow R_{\text{eq}} = \frac{3R}{4R+3}$$

(i) $R = 2\,\Omega$:

$$R_{\text{eq}} = \frac{3R}{4R+3} = \boxed{0.5454\,\Omega}$$

(ii) $R = 4\,\Omega$:

$$R_{\text{eq}} = \frac{3R}{4R+3} = \boxed{0.6315\,\Omega}$$

(iii) $R = 0\,\Omega$:

$$R_{\text{eq}} = \frac{3R}{4R+3} = \boxed{0\,\Omega}$$

3.46

46. Making appropriate use of resistor combination techniques, calculate i_3 and v_x in the circuit of Fig. 3.85.

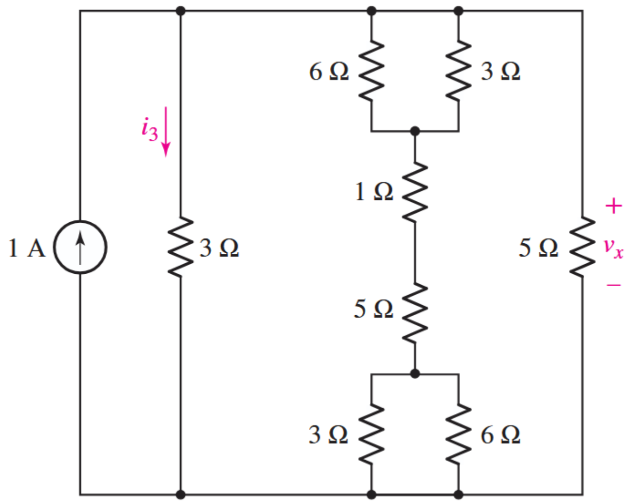


FIGURE 3.85

Combine $3\,\Omega$ and $6\,\Omega$ resistors in parallel:

$$\frac{[3\,\Omega]\cdot[6\,\Omega]}{[3\,\Omega]+[6\,\Omega]} = 2\,\Omega$$

Combine all resistors on the second branch in series:

$$[2\,\Omega]+[1\,\Omega]+[5\,\Omega]+[2\,\Omega] = 10\,\Omega$$

Now combine resistors in all 3 branches in parallel:

$$\frac{1}{R_{eq}} = \frac{1}{[3\,\Omega]} + \frac{1}{[10\,\Omega]} + \frac{1}{[5\,\Omega]} \quad \implies R_{eq} = 1.578\,\Omega$$

Using Ohm's law to find voltage across each branch:

$$V = [1\,\text{A}][1.578\,\Omega] = 1.578\,\text{V}$$

Thus:

$$v_3 = i_3 \cdot [3\,\Omega] = [1.578\,\text{V}] \quad \implies i_3 = \boxed{0.526\,\text{A}}$$

$$\boxed{v_x = 1.578\,\text{V}}$$

3.48

48. Determine the power absorbed by the 15 Ω resistor in the circuit of Fig. 3.87.

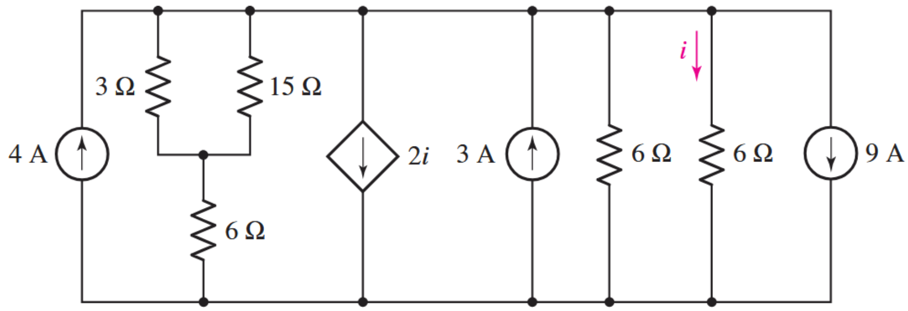
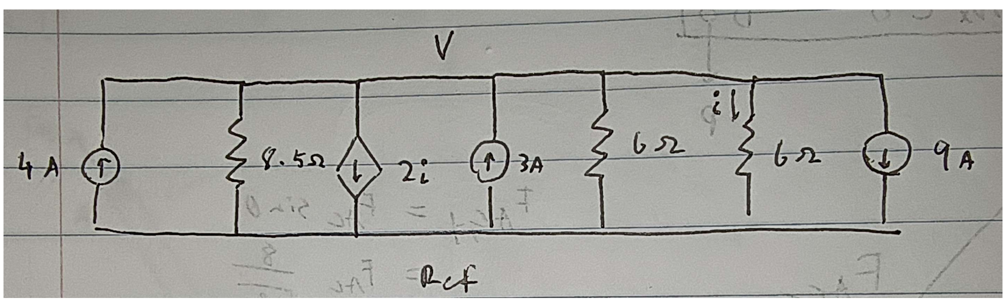


FIGURE 3.87

First simplify the branch with the $3\,\Omega$, $15\,\Omega$, and $6\,\Omega$ resistors:

$$R = \frac{[3\,\Omega] \cdot [15\,\Omega]}{[3\,\Omega] + [15\,\Omega]} + [6\,\Omega] = 8.5\,\Omega$$

Apply Nodal analysis (all elements are in $\|$ so the node voltage is the same across all elements):



KCL at top node of 3 A source:

$$[3\,\mathrm{A}] = 2i - [4\,\mathrm{A}] + [9\,\mathrm{A}] + \frac{V}{[8.5\,\Omega]} + \frac{V}{[6\,\Omega]} + \frac{V}{[6\,\Omega]}$$

$$i = \frac{V}{[6\,\Omega]}$$

Thus, solving for V:

$$3 = \frac{2V}{6} - 4 + 9 + \frac{V}{8.5} + \frac{V}{6} + \frac{V}{6} \quad \Longrightarrow V = -2.55\,\mathrm{V}$$

Using the voltage divider rule on the $8.5\,\Omega$ branch to find v_{15} (which is the same as the $2.5\,\Omega$ parallel combination):

$$v_{2.5} = [-2.55\,\mathrm{V}]\frac{[2.5\,\Omega]}{[8.5\,\Omega]} = -0.75\,\mathrm{V}$$

The power absorbed by the $15\,\Omega$ resistor is then:

$$i_{15} = \frac{[-0.75\,\mathrm{V}]}{[15\,\Omega]} = -0.05\,\mathrm{A}$$

$$P_{15} = [-0.75\,\mathrm{V}] \cdot [-0.05\,\mathrm{A}] = \boxed{0.0375\,\mathrm{W}}$$

Note: The voltage divider rule applies to series elements, where

$$v_x = V \frac{R_x}{R_{\text{total}}}$$

3.54

54. Employ voltage division to assist in the calculation of the voltage labeled v_x in the circuit of Fig. 3.92.

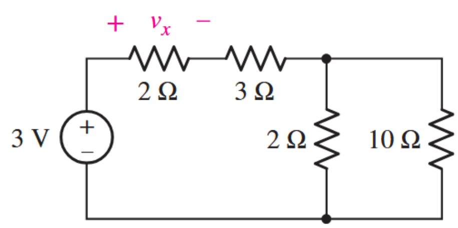


FIGURE 3.92

Combine the $2\,\Omega$ and $10\,\Omega$ resistors in parallel:

$$R_{\|} = \frac{[2\,\Omega] \cdot [10\,\Omega]}{[2\,\Omega] + [10\,\Omega]} = 1.66\,\Omega$$

Use voltage divider rule to find v_x:

$$v_x = [3\,\mathrm{V}] \frac{[2\,\Omega]}{[2\,\Omega] + [3\,\Omega] + [1.66\,\Omega]} = \boxed{0.692\,\mathrm{V}}$$

3.56

56. Employing resistance combination and current division as appropriate, determine values for i_1, i_2, and v_3 in the circuit of Fig. 3.93.

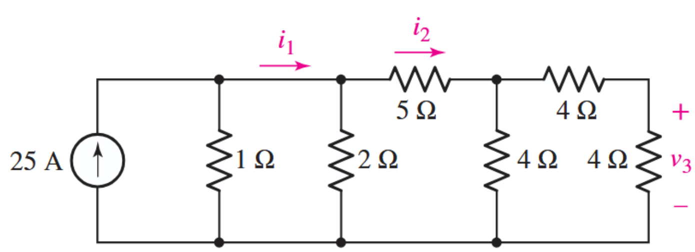


FIGURE 3.93

Finding the equivalent resistance of the circuit (starting right to left):
$4\,\Omega$ and $4\,\Omega$ in series:

$$4 + 4 = 8\,\Omega$$

$8\,\Omega$ and $4\,\Omega$ in parallel:

$$\frac{8 \cdot 4}{8 + 4} = 2.66\,\Omega$$

$2.66\,\Omega$ and $5\,\Omega$ in series:

$$2.66 + 5 = 7.66\,\Omega$$

7.66 Ω and 2 Ω in parallel:

$$\frac{7.66 \cdot 2}{7.66 + 2} = 1.66\,\Omega$$

1.66 Ω and 1 Ω in parallel:

$$R_{\text{eq}} = \frac{1.66 \cdot 1}{1.66 + 1} = 0.624\,\Omega$$

Thus the voltage over each branch is:

$$V = [25\,\text{A}][0.624\,\Omega] = 15.6\,\text{V}$$

Using KCL at the top node of the 1 Ω resistor to find i_1:

$$[25\,\text{A}] = \frac{[15.6\,\text{V}]}{[1\,\Omega]} + i_1 \qquad \Longrightarrow \boxed{i_1 = 9.4\,\text{A}}$$

Using KCL at the top node of the 2 Ω resistor to find i_2:

$$[9.4\,\text{A}] = \frac{15.6\,\text{V}}{[2\,\Omega]} + i_2 \qquad \Longrightarrow \boxed{i_2 = 1.6\,\text{A}}$$

Using the voltage divider rule to find the voltage after the 5 Ω resistor (the equivalent resistance over the three 4 Ω resistors was found to be 2.66 Ω):

$$v = [15.6\,\text{V}]\frac{[2.66\,\Omega]}{[5\,\Omega] + [2.66\,\Omega]} = 5.417\,\text{V}$$

Using voltage divider rule again to find the v_3:

$$v_3 = [5.417\,\text{V}]\frac{[4\,\Omega]}{[4\,\Omega] + [4\,\Omega]} = \boxed{2.708\,\text{V}}$$

3.60

60. The circuit depicted in Fig. 3.97 is routinely employed to model the midfrequency operation of a bipolar junction transistor–based amplifier. Calculate the amplifier output v_{out} if the transconductance g_m is equal to 322 mS.

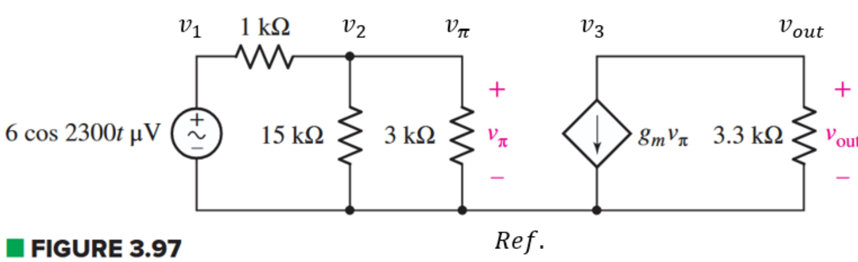


FIGURE 3.97

Applying Nodal analysis at v_2:

$$\frac{v_2 - v_1}{[1 \times 10^3\,\Omega]} = \frac{v_2}{[15 \times 10^3\,\Omega]} + \frac{v_\pi}{[3 \times 10^3\,\Omega]}$$

Since they are parallel, $v_2 = v_\pi$. Also since $v_1 = 6cos(2300t) \times 10^{-6}$ V, we can solve for v_π:

$$\frac{v_\pi - 6cos(2300t) \times 10^{-6}}{[1 \times 10^3\,\Omega]} = \frac{v_\pi}{[15 \times 10^3\,\Omega]} + \frac{v_\pi}{[3 \times 10^3\,\Omega]}$$

$$\implies v_\pi = cos(2300t) \times 10^{-5}\,\text{V}$$

Nodal analysis at v_{out}:

$$-g_m v_\pi = \frac{v_{out}}{[3.3 \times 10^3\,\Omega]}$$

Plugging in values:

$$-[0.322\,s][cos(2300t) \times 10^{-5}\,\text{V}] = \frac{v_{out}}{[3.3 \times 10^3\,\Omega]} \implies \boxed{v_{out} = -1.062cos(2300t) \times 10^{-2}\,\text{V}}$$

3.61

61. With regard to the circuit shown in Fig. 3.98, compute (*a*) the voltage across the two 10 Ω resistors, assuming the top terminal is the positive reference, and (*b*) the power dissipated by the 47 Ω resistor. (*c*) If the maximum rating of the 47 Ω resistor is 0.25 W, is it exceeded by this circuit? Explain.

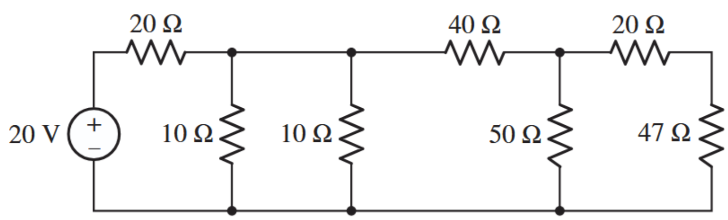


FIGURE 3.98

(a) Finding the equivalent resistance of the circuit (starting right to left) like in problem *3.56*:

20 Ω and 47 Ω in series:

$$20 + 47 = 67\,\Omega$$

67 Ω and 50 Ω in parallel:

$$\frac{67 \cdot 50}{67 + 50} = 24.3589\,\Omega$$

24.3589 Ω and 40 Ω in series:

$$24.3589 + 40 = 64.3589\,\Omega$$

64.3589 Ω and 10 Ω in parallel:

$$\frac{64.3589 \cdot 10}{64.3589 + 10} = 8.6551\,\Omega$$

8.6551 Ω and 10 Ω in parallel:

$$\frac{8.6551 \cdot 10}{8.6551 + 10} = 4.6395\,\Omega$$

Using voltage divider rule to find voltage across the 4.6395 Ω equivalent resistance (which is the same as across the 10 Ω resistors) in series with the 20 Ω resistor:

$$v = [20\,\text{V}]\frac{[4.6395\,\Omega]}{[20\,\Omega] + [4.6395\,\Omega]} = 3.7659\,\text{V}$$

$$\boxed{v_{10\,\Omega} = 3.7659\,\text{V}}$$

(b) Using the voltage divider rule to find voltage across the $24.3589\,\Omega$ equivalent resistance (which is the same as across the $50\,\Omega$ resistor) in series with the $40\,\Omega$ resistor:

$$v = [3.7659\,\mathrm{V}]\frac{[24.3589\,\Omega]}{[40\,\Omega] + [24.3589\,\Omega]} = 1.4253\,\mathrm{V}$$

Finally, one more time using the voltage divider rule to find voltage across the $47\,\Omega$ resistor:

$$v = [1.4253\,\mathrm{V}]\frac{[47\,\Omega]}{[20\,\Omega] + [47\,\Omega]} = 0.9998\,\mathrm{V}$$

Thus, the power dissipated by the $47\,\Omega$ resistor is:

$$P = iv = \frac{v^2}{R} = \frac{[0.9998\,\mathrm{V}]^2}{[47\,\Omega]} = \boxed{0.02126\,\mathrm{W}}$$

(c) The rating is not exceeded since $0.25\,\mathrm{W} > 0.02126\,\mathrm{W}$.